# Network Threats and Mitigation & Physical Security and Risk

**Task 1 : Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).**

To identify five common network threats and understand how they can affect applications and online services.

| No. | Network Threat | Description | Real-World Example |
|---|---|---|---|
| 1 | Phishing | Attackers create fake messages or websites to steal personal information. | A fake Instagram login link sent through WhatsApp to steal login credentials. |
| 2 | DDoS Attack | Attackers flood a server with excessive traffic to make a service unavailable. | A ticket booking website like BookMyShow becoming unavailable due to malicious traffic. |
| 3 | Man-in-the-Middle (MITM) | An attacker secretly intercepts communication between two parties. | An attacker monitoring unencrypted communication when a user accesses an unsafe public Wi-Fi network to use Paytm. |
| 4 | Malware | Malicious software designed to damage systems or steal information. | A user downloads a fake WhatsApp application containing malware that can steal personal data. |
| 5 | Password Attack | Attackers try to steal or guess passwords to gain unauthorized access. | An attacker attempts multiple passwords to access an Instagram account. |

**Task 2 : Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.**

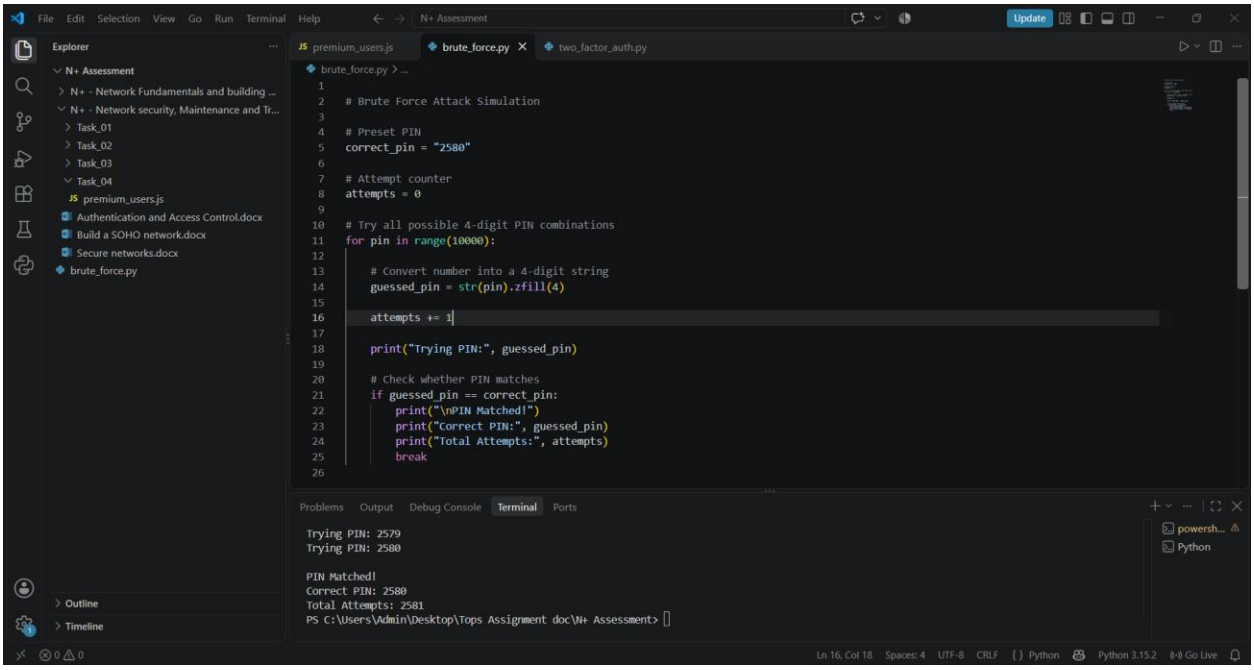
To simulate a brute-force attack by trying all possible combinations of a 4-digit PIN using Python.

**Description:**
The program uses a for loop to generate PIN combinations from 0000 to 9999. It compares each generated PIN with a predefined PIN. A counter records the number of attempts required to find the correct PIN.

**Preset PIN:** 2580

**Output:**

- Correct PIN: 2580
- Total Attempts: 2581
- Status: PIN Matched!



**Task 3 : List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.**

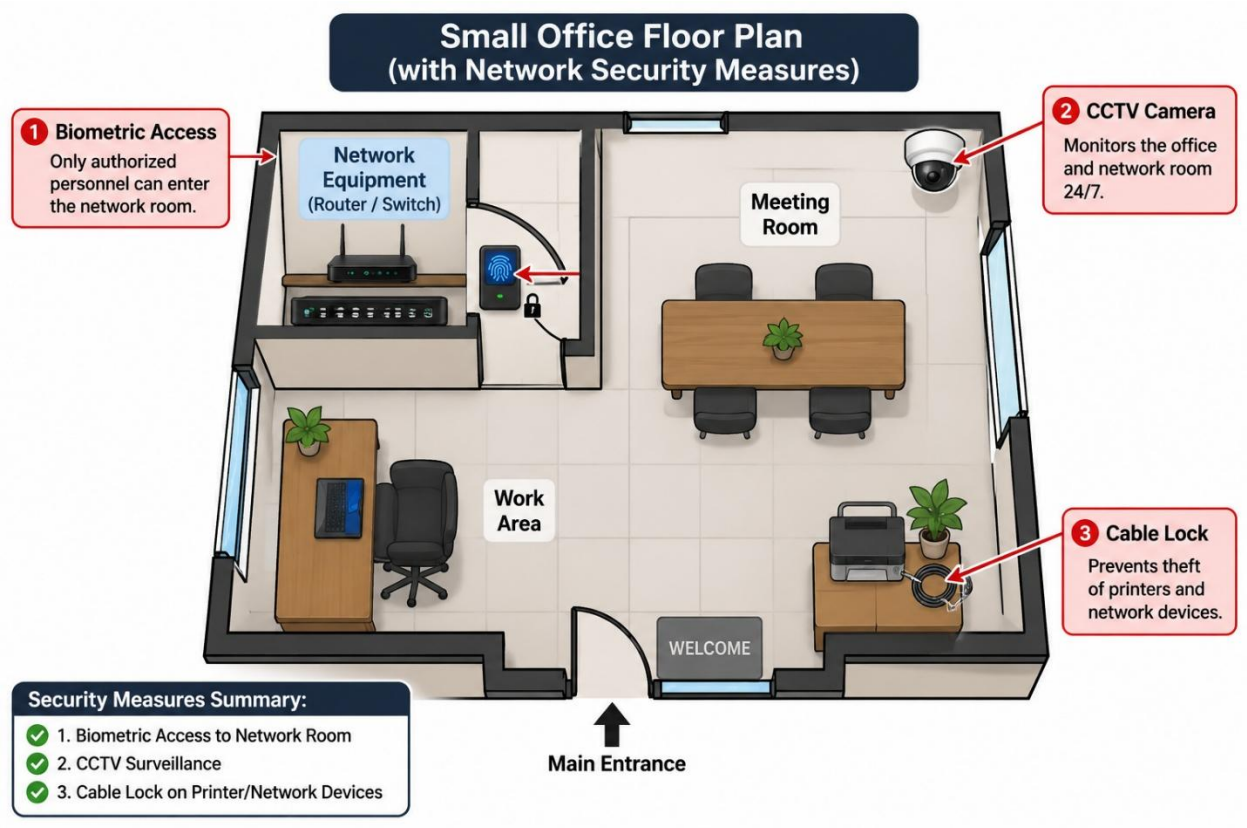
To identify physical security risks in a server room and suggest suitable prevention methods.

| No. | Physical Security Risk | Description | Mitigation |
|---|---|---|---|
| 1 | Unauthorized Access | An unauthorized person may enter the server room and damage or steal network equipment. | Install biometric access control and allow entry only to authorized staff. |
| 2 | Fire | Electrical faults or overheating can cause a fire and damage servers. | Install smoke detectors, fire alarms, and suitable fire suppression systems. |
| 3 | Power Failure | Sudden power failure can shut down servers and interrupt services. | Install UPS and backup generators to provide continuous power. |
| 4 | Water Leakage | Water from pipes or cooling systems can damage servers and electrical equipment. | Install water leakage sensors and keep equipment away from water pipes. |

**Conclusion:**

Physical security is important for protecting server rooms and network equipment. Proper access control, fire protection, backup power, and water leakage detection help prevent damage and service interruptions.

**Task 4 : Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).**




### 1. Biometric Access Control

A fingerprint scanner is installed at the entrance of the network room. Only authorized employees can access the room where the router and network switch are kept.

### 2. CCTV Surveillance

CCTV cameras are installed to monitor the office and network room 24/7. They help detect suspicious activities and provide video recordings for security purposes.

### 3. Cable Locks

Cable locks are used to secure devices such as printers and network equipment. They help prevent theft or unauthorized removal of devices.

**Task 5 : Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.**

To create a checklist of seven security actions that a network administrator should follow to protect a college computer lab.

**Security Checklist:**

1. **Install and Update Antivirus:** Install updated antivirus software on all lab computers to detect and remove malware.
2. **Configure Firewall:** Enable Windows Defender Firewall on every computer to block unauthorized network connections.
3. **Use Strong Passwords:** Create strong passwords for administrator accounts and restrict access to authorized users only.
4. **Secure Wi-Fi Network:** Protect the college Wi-Fi network using WPA2 or WPA3 encryption and a strong password.
5. **Install CCTV Cameras:** Install CCTV cameras in the computer lab to monitor activities and prevent unauthorized access.
6. **Restrict Physical Access:** Keep the server room and networking equipment locked. Allow access only to authorized staff.
7. **Perform Regular Backups and Updates:** Regularly update operating systems and applications and maintain backups of important college lab data.

**Review and Customization:**
The checklist was reviewed and modified according to a college computer lab environment. It focuses on protecting student computers, network devices, Wi-Fi connections, server rooms, and important academic data.